

Latest MCU Development Boards Make Prototyping Easier



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Embodied design prototyping has never been so easy before," exclaims Darshan Virupaksha, systems engineer, Altium Innovations, and co-organiser, IoTBLR. He says, "Makers and do-it-yourself (DIY) enthusiasts have a plethora of options to suit their needs. Today, the ecosystem offers easy access to development boards and the most important feature is software support."

The prototyping scenario has indeed seen an escalation and one of the biggest factors is the range of microcontroller (MCU) development boards out there in market. From low-cost, entry-level boards to powerful, application-specific ones, we try to help you buy the right board by focusing on different categories, new features, general selection criteria and boards that have been in the spotlight.

General, application- and industrial-specific boards

Quite a few development boards in the market today are general MCU-specific, such as

STM32 discovery kits, MSP430 from Texas Instruments (TI) or Xplained boards from Atmel, and these are available from semiconductor manufacturers themselves, informs T. Anand, managing director, Knewtron.

Local companies design some of the economically available boards. Anand says, "In both cases, MCU development boards come in two flavours; one is the MCU evaluation board, where the focus is on evaluating MCU capabilities."

"Second category is application- or feature-specific, where the board is designed for the purpose of evaluating certain features of the MCU and is usually focussed on some application area," he adds.

Broadly, there are three different categories that today's MCU boards fall under.

Generic multi-purpose boards. Today, every MCU manufacturer like Microchip, NXP, Cypress, TI and Atmel provides an evaluation kit compatible with their MCUs. Kanishka Shah, co-founder of Entesla, says, "International third-party development board

Some MCU
development boards



MediaTek LinkIt ONE development platform

ARM mbed IoT Device Platform

TI's MSP430
LaunchPad